2020-Summer Paper 3 – Subspecialities of Urology

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Q1. Etio-pathogenesis, diagnosis and management of Recto-urethral fistula?

Type: LAQ (25 marks)

Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra and the distal rectum. Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

Aetiology

Group	Causes
Congenital	Anorectal malformation in boys (rectobulbar or rectoprostatic fistula)
Surgical (commonest acquired)	**RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate), open prostatectomy
Radiation, energy	EBRT (external beam radiotherapy) plus brachytherapy (commonest), brachytherapy alone; cryotherapy; HIFU (high-intensity focused ultrasound), up to about 9% after salvage HIFU
Others	Pelvic fracture, penetrating injury; Crohn disease; advanced rectal or prostate cancer

Pathogenesis

- **Surgical:** an unrecognised rectal tear near the apex during posterior (Denonvilliers) dissection. Urine leaks into it, an abscess forms and the tract epithelialises. Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%.
- **Radiation or ablation:** obliterative endarteritis makes tissue hypoxic and fibrotic; necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

Classification: Martini (after prostate cancer treatment)

Axis	Category	Definition
Stage (size on CT (computed tomography) or cystography)	I / II / III	≤1.5 cm / >1.5 cm / any size with sphincter damage
Grade (tissue)	0	Surgical injury, no prior non-surgical treatment
	1	Prior radiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation

- **Tissue quality (grade), not size, decides the repair.**

Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding; recurrent UTI (urinary tract infection), urosepsis.
- Coexisting SUI (stress urinary incontinence); DRE (digital rectal examination) may feel the defect.

Diagnosis

- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; size and site.
- Upper-tract imaging; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later.

Prevention

- At RP: a recognised tear without gross spillage → two-layer closure, air test, **omental interposition**, drain, antibiotics; colostomy if irradiated.
- Radiotherapy: **hydrogel rectal spacer** (randomised trial: late rectal toxicity 2.0% vs 7.0%).

Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close in **tension-free, non-overlapping layers with vascularised interposition tissue**.
- **Conservative:** no sepsis and no faecaluria → catheter, low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; a **diverting colostomy usually comes first**.

Approach	Essentials	Strengths	Weaknesses
Perineal (Young; Goodwin)	Separate and close urethra and rectum; **gracilis flap** between (or dartos; buccal graft)	High closure; most stomas reversed	Urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure	Avoids scarred plane; spares sexual function	No flap; faecal incontinence up to 20%
Abdominal / robotic	Transvesical or extravesical; omental flap	Large or irradiated fistulae	Morbid; small robotic series

Approach	Essentials	Strengths	Weaknesses
Salvage	Cystoprostatectomy, ileal conduit ± permanent colostomy	Unreconstructable irradiated field	Major surgery

```
RUF → sepsis control, catheter ± colostomy, biopsy
  small, non-irradiated → conservative ~12 weeks
  persistent, grade 0 → perineal (gracilis) or York-Mason
  irradiated or large → abdominal/robotic (omentum) or perineal (gracilis)
  unreconstructable → urinary + faecal diversion
Congenital → newborn colostomy → posterior sagittal anorectoplasty
```

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it in the sphincter complex.

Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, stricture, ED (erectile dysfunction), wound infection.

Diagram to draw: sagittal pelvis showing the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated need imported tissue or diversion.

Q2. Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?

Type: LAQ (25 marks)

Development (summary; for detail see 2020-Summer Paper 1, Q2)

- **Weeks 4–6:** the ureteric bud arises from the mesonephric (Wolffian) duct. The common excretory duct is absorbed into the UGS (urogenital sinus), so ureter and duct open separately. The ureteric orifice moves cranially and laterally; the ejaculatory duct ends caudally in the prostatic urethra.
- **Bud position (Mackie–Stephens):** a bud arising too low reaches the UGS early → high, lateral orifice → short tunnel → **VUR (vesicoureteral reflux)**, often with a dysplastic kidney. A bud arising too high → caudal, medial, ectopic orifice.
- **Weigert–Meyer** (duplex): lower-pole orifice lateral, refluxes; upper-pole caudal-medial, ectopic.

Clinical anatomy of the VUJ (vesico-ureteric junction)

- **Course:** juxtavesical ureter in **Waldeyer's sheath** → oblique intramural segment through the detrusor hiatus → submucosal tunnel → orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). **Narrowest part** of the ureter (about 3–4 mm): commonest stone impaction site.
- **Trigone:** superficial trigone (ureteric longitudinal muscle, to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Base is the interureteric ridge.
- **Antireflux mechanism:**
  - **Passive flap valve:** filling compresses the submucosal ureter against firm detrusor backing.
  - **Paquin:** tunnel length to diameter **5:1** in normal VUJs vs 1.4:1 in refluxing ones.
  - **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
  - **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal) → stadium → horseshoe → **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

Grade 3 VUR (International Reflux Study)

- Mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.

VUR in adults

- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI (urinary tract infection) coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).
- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD (chronic kidney disease).

Evaluation

- **VCUG (voiding cystourethrogram):** grade, bladder, diverticulum, outlet.
- US (ultrasound) and **DMSA (dimercaptosuccinic acid) scan** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected.
- **UDS (urodynamics):** compliance, voiding pressure, residual. Cystoscopy; blood pressure, creatinine, urine protein.

Management

1. Treat secondary causes first (relieve BOO; CIC (clean intermittent catheterisation) and antimuscarinic for neurogenic bladder), then reassess.

2. Primary grade 3 VUR:

Situation	Management
Incidental; no UTI, no scars	Observe: timed voiding, bowel care, prompt UTI treatment
Recurrent febrile UTI, new scars, falling function	**Correct the reflux**
Woman planning pregnancy with scars or prior pyelonephritis	Consider correction before pregnancy
Poorly functioning scarred kidney with infection or hypertension	Nephrectomy

- Low-dose prophylaxis selectively; adult evidence is limited.

3. Correction:

- **Endoscopic injection** (Dx/HA (dextranomer/hyaluronic acid)): day case, repeatable, but less effective in adults (see 2020-Summer Paper 4, Q3c).
- **Ureteroneocystostomy** (open, laparoscopic or robotic): gentle handling, tension-free, straight course, tunnel **5:1**.

Technique	Type	Note
Lich–Gregoir / **RALUR-EV** (robot-assisted extravesical)	Extravesical detrusorotomy	Suits adults; no cystotomy; stay close to ureter (nerves)
Politano–Leadbetter	Intravesical, new hiatus	Long tunnel
Cohen cross-trigonal	Intravesical	Over 98% success; later retrograde access hard

- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.

Complications and follow-up

- Persistent or contralateral reflux, obstruction, leak, retention after bilateral extravesical repair.
- US at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

Diagram to draw: oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.

> **Key points to remember:**
>
> - Bud too low → lateral orifice → short tunnel → VUR.
> - Adult intramural ureter 1.2–2.5 cm; Paquin 5:1.
> - In adults the scar, not the reflux, carries the risk; exclude secondary causes first.
> - Correct for recurrent pyelonephritis, new scars or pregnancy plans.

Q3a. Management of lower ureteric stone of size less than 1 cm in pregnancy?

Type: SAQ (10 marks)

Background

- About 1 in 200 to 1 in 1500 pregnancies; **renal colic is the commonest non-obstetric cause of admission**. Mostly second or third trimester; mainly **calcium phosphate** stones. Linked with preterm delivery and pre-eclampsia.
- Physiological hydronephrosis (up to 90% by third trimester) mimics obstruction.

Diagnosis

- Urine culture always; blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; Doppler RI (resistive index) rises in acute obstruction.
- Second line: **MRI (magnetic resonance imaging)**, stone as a filling defect; **low-dose CT (computed tomography)**. Total fetal dose under **50 mGy**.

Management

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Lower ureteric stone <1 cm
 ├─ sepsis / solitary kidney / renal failure → urgent stent or nephrostomy
 ├─ pain controlled, no infection → observation (50–80% pass)
 └─ failed observation → ureteroscopy + laser
                        or stent/nephrostomy until delivery
```

- **Conservative:** hydration, paracetamol, opioids. **Avoid NSAIDs (non-steroidal anti-inflammatory drugs)** (ductus arteriosus closure in third trimester). **MET (medical expulsive therapy)** with tamsulosin is off-label.
- **URS (ureteroscopy) with holmium laser:** first-line intervention for a non-complex stone in any trimester; ideal for a distal stone. Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (US guidance, shield under the patient), fetal monitoring. Third-trimester URS may raise preterm labour risk.
- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents every **4–6 weeks**, nephrostomy about every 3 weeks; definitive treatment after delivery.
- **SWL (shock wave lithotripsy) contraindicated**. Second trimester is the safest window.

Diagram to draw: the flowchart above.

> **Key points to remember:**
> - US first; MRI or low-dose CT second.
> - Observation first; avoid NSAIDs.
> - URS with laser is definitive; SWL contraindicated.
> - Stents encrust: change every 4–6 weeks.

Q3b. Type 2 Renal tubular acidosis?

Type: SAQ (10 marks)

Definition

- **Proximal RTA (renal tubular acidosis):** failure of the proximal tubule to reclaim bicarbonate, causing **hyperchloraemic, normal anion gap metabolic acidosis**.

Pathophysiology

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–HCO_3 cotransporter).
- **Self-limiting:** bicarbonate wasting → plasma HCO_3^- falls → filtered load drops below the lowered threshold → the intact distal tubule copes → **steady state at about 14–18 mEq/L with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

Causes

Isolated	Generalised (Fanconi syndrome)
Carbonic anhydrase inhibitors (acetazolamide, **topiramate**)	**Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia
Carbonic anhydrase II deficiency (mixed)	**Ifosfamide, cisplatin**, aminoglycosides, valproate
	Myeloma, amyloidosis, lead

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

Clinical features

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.
- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

Diagnosis

- Normal anion gap acidosis, low potassium; urine can acidify below 5.5 (unlike distal RTA).
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%**.
- Look for Fanconi features; drug history.

Treatment

- Treat the cause.
- **Large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium.
- Thiazide; phosphate and vitamin D for rickets.

	Type 1 (distal)	Type 2 (proximal)	Type 4
Urine pH in acidosis	Always >6	<5.5 at steady state	Low
Potassium	Low	Low	High
Stones	Common (up to 70%)	Uncommon	Uncommon

Diagram to draw: proximal tubular cell with carbonic anhydrase.

Key points to remember:

- Bicarbonate wasting; HCO_3^- 14–18 mEq/L; urine pH <5.5 at steady state.
- Usually Fanconi; ifosfamide, cisplatin.
- Fractional HCO_3^- excretion >15%; stones uncommon.

Q3c. Auto-transplantation of Kidney?

Type: SAQ (10 marks)

Definition

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) repair** if needed. First reported in 1963 for proximal ureteric injury.

Indications

Group	Examples
Ureteric (commonest)	Long-segment loss or stricture; failed repeated repairs; **RPF (retroperitoneal fibrosis)** when ureterolysis fails
Vascular	Complex renal artery aneurysm or branch disease, AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome
Tumour	Bench partial nephrectomy in a solitary kidney

- Chosen when the **contralateral kidney is absent or poor**, or ileal ureter is unsuitable (creatinine >2 mg/dL, bowel disease).
- **Contraindications:** poor function, severe iliac atherosclerosis, unfit patient.

Procedure

```
CT (computed tomography) angiography, renogram, ureteric study
→ harvest as for live donor nephrectomy (open/laparoscopic/robotic),
  maximal vessel length, preserve ureteric blood supply
→ cold perfusion on bench ± ex vivo repair
→ Gibson incision, extraperitoneal iliac fossa
→ renal vein to external iliac vein (end-to-side)
→ renal artery to external iliac (end-to-side) or internal iliac (end-to-end)
→ drainage: pyelovesicostomy (± Boari flap), ureteroneocystostomy,
  or native ureter left in continuity
→ stent, catheter, drain
```

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; robotic autotransplantation emerging.

Complications

- Arterial or venous **thrombosis**, bleeding, **renal unit loss 5–10%**, urine leak, stricture, lymphocele, infection.
- Major (Clavien–Dindo ≥III) complications about 15–20%.

Outcome

- Long-term success **about 90%**. Doppler US (ultrasound) early; renogram and blood pressure later.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

Key points to remember:

- For long ureteric loss when other repairs fail and the other kidney is poor.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

Q3d. Indication, procedure and complications of Uretero-Calycostomy?

Type: SAQ (10 marks)

Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

Procedure (open, laparoscopic or robotic)

```
Flank position; pre-placed double-J stent
→ mobilise ureter with generous periureteral tissue
→ mobilise kidney; partial lower-pole nephrectomy
   to expose the calyx widely (not simple nephrotomy)
→ close the pelvic outlet; spatulate ureter laterally
→ apex stitch to lateral calyx wall; second at 180°
→ interrupted sutures, all placed then tied
→ capsule closed over cut surface, away from anastomosis
→ cover with perinephric fat, peritoneum or omentum; drain
```

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.
- Stent out at 4–6 weeks; renogram later. Later endoscopy needs **antegrade upper-pole access**.

Complications

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, renal unit loss.
- Success about 70% in the largest (mostly redo) series; better in small paediatric series.

Also know: alternatives are redo pyeloplasty, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy.

Diagram to draw: kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection.
> - Stricture is the main failure; success about 70% in redo series.

Q3e. Management of bilateral Wilms Tumor?

Type: SAQ (10 marks)

Introduction

- Synchronous bilateral WT (Wilms tumour) in **5–7%** = **COG (Children's Oncology Group) stage V**; each kidney is staged separately.
- **Nephrogenic rests in over 90%**; syndromes: **WAGR (Wilms, aniridia, genitourinary anomalies, retardation)** and Denys–Drash (about 50% risk), Beckwith–Wiedemann.
- **Goal:** cure while preserving nephrons. ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.

Evaluation

- US (ultrasound) → contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen; chest CT; coagulation; syndrome review.
- **No upfront biopsy and no upfront nephrectomy.**

Management (COG AREN0534)

- VCR = vincristine, AMD = actinomycin D, DOX = doxorubicin, NSS = nephron-sparing surgery, RT = radiotherapy.

```
VCR + AMD + DOX × 6 weeks (no biopsy)
→ imaging at week 6
   ├─ resectable → bilateral NSS
   └─ not → 6 more weeks → surgery by week 12
        poor response → bilateral OPEN biopsy
→ further therapy by each kidney's stage and histology ± flank RT
```

- **Why 12 weeks:** in NWTS-4 (National Wilms Tumor Study-4) prolonged chemotherapy delayed surgery and some tumours became anaplastic.
- **Why open, bilateral biopsy:** anaplasia differs between kidneys in about 4 of 5; non-shrinking tumour is often differentiated and cured by excision.
- **SIOP (International Society of Paediatric Oncology):** same principle with VCR + AMD if non-metastatic.

Surgery

- Partial nephrectomy or **enucleation** (positive margins 23–36%, good outcome with flank RT). Expert centres achieve bilateral NSS in over 90%.
- Bilateral nephrectomy and dialysis only for refractory tumour; **transplant after 1–2 years**.

Outcome and follow-up

- AREN0534: **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%** (vs 56% and 80.8% on NWTS-5).
- Lifelong renal function, proteinuria and blood pressure checks; late relapse possible.

Diagram to draw: AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
> - Stage V; no upfront nephrectomy or biopsy.
> - Three drugs; reassess at 6 weeks; surgery by 12 weeks.
> - Bilateral NSS to avoid ESRD.

Q3f. Inguinal Block Dissection?

Type: SAQ (10 marks)

Taken as ILND (inguinal lymph node dissection), mainly for penile cancer. For VEIL see 2019-Summer Paper 4, Q3d; for nodal strategy see 2017-Summer Paper 2, Q1.

Indications

- Penile SCC (squamous cell carcinoma): **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB (dynamic sentinel node biopsy) is positive or unavailable. Also vulval cancer, melanoma.
- **Within 3 months**; aim for **at least 8 nodes per side**. Early ILND: 3-year disease-specific survival about 84% vs 35% if delayed (Kroon).
- Bulky or fixed nodes: chemotherapy first.

Surgical anatomy

- **Femoral triangle:** inguinal ligament, sartorius (lateral), adductor longus (medial); floor fascia lata.
- **Superficial nodes** (4–25) in **Daseler's five zones**; sentinel nodes mostly **superomedial**.
- **Deep nodes** (1–3) medial to femoral vein; **Cloquet's node** highest. Spread is stepwise, without skips.

Templates

Template	Extent
Radical	Superficial + deep; saphenous often ligated; sartorius transposed
Modified (Catalona)	Nothing lateral to femoral artery or below fossa ovalis; saphenous spared; positive frozen section → radical
Superficial	Above fascia lata only

Technique (open radical)

- Frog-leg; incision **2 cm below and parallel** to the ligament.
- Flaps **deep to Camper's fascia** (skin capillaries).
- Divide saphenous at apex; open fascia lata; clear along femoral vessels; **never lateral to the femoral artery** (femoral nerve).
- **Sartorius transposition**; suction drains until <30 mL/day.
- Pelvic dissection if more than 2 inguinal nodes or ENE (extranodal extension).

Complications

- **Wound infection, flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis).
- Reduced by correct flap plane, modified template, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.

Diagram to draw: femoral triangle with Daseler's zones.

> **Key points to remember:**
>
> - Ligament, sartorius, adductor longus.
> - Flaps deep to Camper's; stay medial to femoral artery.
> - Within 3 months; at least 8 nodes per side.

Rapid Revision Sheet – 2020-Summer Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? (LAQ)

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated need imported tissue or diversion.

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- Bud too low → lateral orifice → short tunnel → VUR.
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- In adults the scar, not the reflux, carries the risk; exclude secondary causes first.
- Correct for recurrent pyelonephritis, new scars or pregnancy plans.

Q3a. Management of lower ureteric stone of size less than 1 cm in pregnancy? (SAQ)

- US first; MRI or low-dose CT second.
- Observation first; avoid NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change every 4–6 weeks.

Q3b. Type 2 Renal tubular acidosis? (SAQ)

- Bicarbonate wasting; HCO_3^- 14–18 mEq/L; urine pH <5.5 at steady state.
- Usually Fanconi; ifosfamide, cisplatin.
- Fractional HCO_3^- excretion >15%; stones uncommon.

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- For long ureteric loss when other repairs fail and the other kidney is poor.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

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- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection.
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- Stage V; no upfront nephrectomy or biopsy.
- Three drugs; reassess at 6 weeks; surgery by 12 weeks.
- Bilateral NSS to avoid ESRD.

Q3f. Inguinal Block Dissection? (SAQ)

- Ligament, sartorius, adductor longus.
- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.